DEPARTMENT OF AGRICULTURE IV-A · DEPDev IV-A

Food Insecurity Risk Forecast

Batangas · Q4 2025 (Oct-Dec)

Generated 29 September 2026 from aiPHeed model output.

> **What this number is**
>
> The modelled probability that this area falls in the HIGH class of the composite food stress label. **It is an indicative relative risk estimate, not an official food insecurity classification**, and it is published at the Maximum Reliable Lead Time the model established empirically.

Headline

FOOD INSECURITY RISK PROBABILITY

0.485

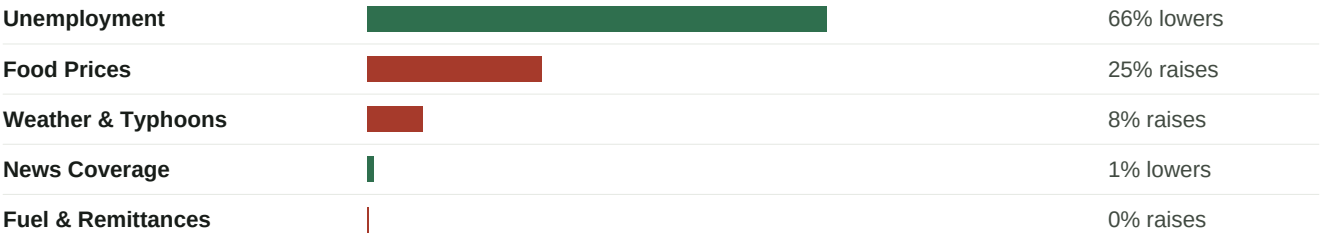
LOW · quarter-on-quarter +0.0000

Risk class	**LOW**
Rank in CALABARZON	**3 of 5**
Population (2020 CPH)	**2,908,494**
Poverty incidence (PSA 2021)	**7.1%**
Articles analysed	**0**

Forecast horizon

Same-quarter estimate. No forward lead time met the 75% reliability threshold of Section 3.4.4, so none is published.. Indicative relative risk only. Walk-forward accuracy did not reach 75% at any forward horizon, so this output is reported as a same-quarter estimate and not as a forecast.

What is driving the figure



Red bars raise the predicted shortfall, **green** lower it. A large bar is not automatically bad news -- share of the explanation and direction of effect are separate things.

For Batangas in 2025-Q4, the model's prediction breaks down as: Unemployment 66%, Food Prices 25%, Weather & Typhoons 8%, News Coverage 1%, Fuel & Remittances 0%. Unemployment is the largest single contributor, holding the score down. Food Prices clears the 20% line while raising the score. Unemployment clears the 20% line while lowering it. Shares are SHAP contributions to the province's food insecurity risk probability, recomputed every quarter.

Municipal distribution

Municipal values reweight the province figure by poverty incidence (60%) and population density (40%). **They are not independent municipal forecasts** -- every municipality in a province rises and falls with it.

#	City / Municipality	Class	Population	Poverty	Index
1	**Sto. Tomas**	City	218,500	7.1%	0.3291
2	**Nasugbu**	Municipality	136,524	9.4%	0.3151
3	**Taal**	Municipality	61,460	7.1%	0.3079
4	**Malvar**	Municipality	64,379	7.1%	0.2972
5	**Tanauan**	City	193,936	7.1%	0.2839
6	**Bauan**	Municipality	90,819	7.1%	0.2739
7	**Talisay**	Municipality	46,238	7.1%	0.2678
8	**Mataasnakahoy**	Municipality	30,621	7.1%	0.2601
9	**San Jose**	Municipality	79,868	7.1%	0.2546
10	**San Pascual**	Municipality	69,009	7.1%	0.2416
11	**Santa Teresita**	Municipality	21,559	7.1%	0.2380
12	**Padre Garcia**	Municipality	51,853	7.1%	0.2310
13	**Mabini**	Municipality	50,858	7.1%	0.2211
14	**Alitagtag**	Municipality	26,819	7.1%	0.2154
15	**San Nicolas**	Municipality	23,908	7.1%	0.2130
16	**Balete**	Municipality	24,055	7.1%	0.2040
17	**Balayan**	Municipality	95,913	7.1%	0.1965
18	**San Luis**	Municipality	36,172	7.1%	0.1934
19	**Lemery**	Municipality	93,186	7.1%	0.1933
20	**Ibaan**	Municipality	58,507	7.1%	0.1933
21	**Agoncillo**	Municipality	39,101	7.1%	0.1871
22	**Calaca**	City	87,361	7.1%	0.1852

#	City / Municipality	Class	Population	Poverty	Index
23	**Lian**	Municipality	56,280	7.1%	0.1824
24	**Cuenca**	Municipality	36,235	7.1%	0.1720
25	**Laurel**	Municipality	43,210	7.1%	0.1705
26	**Tingloy**	Municipality	19,215	7.1%	0.1681
27	**Calatagan**	Municipality	58,719	7.1%	0.1678
28	**Rosario**	Municipality	128,352	7.1%	0.1666
29	**Tuy**	Municipality	46,519	7.1%	0.1596
30	**Taysan**	Municipality	40,146	7.1%	0.1537
31	**San Juan**	Municipality	114,068	7.1%	0.1527
32	**Lipa**	City	372,931	5.1%	0.1459
33	**Batangas City**	City	351,437	5.8%	0.1425
34	**Lobo**	Municipality	40,736	7.1%	0.1353

News evidence

No relevant articles were geocoded to this subject for the quarter.

Method

The model predicts, for each CALABARZON province, the probability that the province falls in the HIGH class of the composite food stress label -- SWS quarterly hunger prevalence combined with the province's food CPI deviation from the regional mean, thresholded at the global median. Inputs are the bias-corrected Food Stress Sentiment Index and its lags and acceleration, the five bilingual trigger proportions, and government indicators from PSA, BSP, DOE, PAGASA and PhilRice.

Measure	Value
Maximum Reliable Lead Time	0 quarter(s)
Accuracy	0.6000
Precision (weighted)	0.9200
Recall (weighted)	0.6000
F1 score (weighted)	0.6762
AUC-ROC	0.9444
F1 target (>= 0.75)	FAIL
AUC-ROC target (>= 0.80)	PASS
Features in the model	12
Baseline: aipheed lgbm combined	0.6727
Baseline: full feature logistic	0.5455
Baseline: lgbm nlp only	0.6182
Baseline: lgbm psa only	0.7273
Baseline: naive persistence	0.8000
Baseline: nlp only logistic	0.5455
Baseline: psa only logistic	0.5818

Accuracy must be read against its baselines. The comparison models above are evaluated on the same walk-forward folds at the same forecast gap, so the figures are directly comparable. Naive persistence in particular is a demanding baseline on an autocorrelated label: a model that does not clearly beat it is not earning its complexity.

Limitations

Indicative risk, not official classification

Outputs are relative ordinal estimates of potential risk. They are not definitive determinations of food insecurity status and must not be read as official government assessments or policy recommendations.

Proxy label, not a direct measurement

No publicly accessible province-quarter food insecurity label exists in the Philippines. The training target is a composite of SWS hunger prevalence, which has no province-level resolution, and PSA food CPI deviation, which reflects supply-side price conditions rather than household hunger experience directly.

Municipal figures are disaggregated, not modelled

Municipal values reweight the province figure by poverty incidence (60%) and population density (40%). Poverty values are largely province means rather than municipal Small Area Estimates, so within-province ordering is indicative only.

Reviewed before publication

Province-quarter figures pass through analyst review. A withheld figure is excluded from this report and from the regional average.

Scope of this assessment

This assessment reports findings only. It does not recommend interventions: no validated mapping exists between this indicator and a response protocol. Pair it with the Department's own standing guidance.